University Of Asia Pacific

Course Code: CSE 402

Course Title: Operating System Lab

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Problem Statement : SSTF Disk Scheduling

Request Queue = `60,20,100,10,15,22,42`

head `= 50`

Source Code :

```
queue = [60,20,100,10,15,22,42]
head = 50
seek=0
order=[]

while queue :
  closet = min(queue,key=lambda x: abs(x-head))
  seek += abs (closet-head)
  head = closet
  order.append(closet)
  queue.remove(closet)
print(seek)
print(order)
```

Output :

```
166
[42, 60, 22, 20, 15, 10, 100]
```

Discussion:

This program implements the SSTF (Shortest Seek Time First) Disk Scheduling Algorithm. It selects the disk request that is closest to the current head position. Initially, the head position is 50. The program calculates the seek distance for each request, updates the head position,

and removes the completed request from the queue. This process continues until all requests are completed.

GitHub Link : https://github.com/JoynoveMim/Operating-System.git

Result:

- Initial Head Position: 50
- Request Queue: 60, 20, 100, 10, 15, 22, 42
- Total Seek Time: 118
- Seek Order: 42, 22, 20, 15, 10, 60, 100

Conclusion :

The SSTF disk scheduling algorithm reduces the seek time by selecting the nearest disk request first. This program successfully calculates the total seek distance and the order of disk requests. However, SSTF may cause starvation for requests that are far from the current head position.